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MILITARY SPECIFICATION

SWITCH, FLOAT, AIRCRAFT FUEL LELEL GNERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers an aircraft fuel level float switch utilized with aircraft fuel tanks using hydrocarbon fuel.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

F-D-680	Dr Cleaning Solvent
QQ-C-320	Chromium Flating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-S-735	Standard Test Fluids; Hydrocarbon

Military

MIL-P-116	Reservation, Methods of
MIL-W-5086	wire, Electrical, 6oo-volt, Copper, Aircraft
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-J-5161	Jet Fuel, Referee
MIL-E-5Z72	Environmental Testing, Aeronautical and Associated Equipment. General Specification for
MIL-P-5315	Packing, Preformed, Hydrocarbon Fuel Resistant
MIL-G-5572	Gasoline, Aviation; Grades 80/87, 91/96, 100/130, 115/45
MIL-J-5624	Jet Fuel, Grades JP-4 and JP-5
MIL-P-6906	Plates, Identification

FSC 2915

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MIL-F-7024	Fluids, Calibrating, for Aircraft Fuel System Components
MIL-P-71 05	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-F-8615	Fuel System Components; General Specification for
MIL-N-25027	Nut, Self-Locking, 250°F, 450°F, and 800°F 125 KSI FTU, 60 KSI FTU and 30 KKKSI FTU
MIL-C-26482	Connectors, Electric, Circular, Miniature Quick Disconnect
MIL-C-26500	Connectors, General Purpose, Electrical, Miniature, Circular, Environment Resisting, 200°C Ambient Temperature
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

No, 595	Odors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-43	Specifications and Standards, Order of Precedence for Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
M320995	Wire, Lock
MS29513	Packing. Preformed, Hydrocarbon Fuel Resistant, O-Ring
M33540	Safety Wiring, General Practices for
MS33586	Metals, Definition of Dissimilar
MS33588	Nuts and Plate Nuts, Self-Locking, Aircraft Design and Usage Limitation of
MS33656	Fitting End, Standard Dimensions for Flared Tube Connections and Gasket Seal

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Air Force-Navy Aeronautical

AN3179	Inductor, 30 Volt Laboratory Test
AND10049	Bosses - Fluid Connection, Internal Straight Thread
AND10050	Bosses, Straight Dimensions for Gasket Seal Straight Thread

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 438	Age Controls for Synthetic Rubber Parts

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific perocurementfuctions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. Committee on Extension to the Standard Atmosphere

U.S. Standard Atmosphere 1962

(Application for copies of above publication should be addressed to the Superintendent of Documents, Government Printing Office, Washington D. C., 20402.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproducticn testing.

3.2 Unless otherwise specified in the contract or order, no data are r ed by this specification or any of the documents referenced in section 2 herein see 6.2).

3.3 Model specification.- When required, a model specification conforming to this specification format and the appendix of MIL-F-8615 shall be submitted for approval.

3.4 Materials.- Materials and processes used by the manufacturer of float switches shall be suitable for the purpose, and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used, provided the specifications are released by the Services and contain provision for adequate tests. The use of contractors specifications shall not constitute waiver of Government inspection. All materials used in the component shall be sufficiently resistant to fluids conforming to MIL-J-5161; MIL-F-7024, type II; MIL-J-5624; MIL-G-5572; and TT-S-735, of aromatic content from 0 to 30 percent, to assure satisfactory operation as herein defined.

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3.4.1 Metals.- All metals used in the construction of float switches that are not of a corrosion-resi stant types shall be suitably protested to re sist corrosion during the normal sevice life of the switch. The use of di ssimilar metals, e specially brass, copper, or steel, in contact with aluminum or alumimum alloy, shall be avoided where practicable. Dissimilar metals are defined in MS33S86. Magnesium shall not be used. Neither brass nor copper shall be used in contact with fuel.

3.4.2 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.4.2.1 Standard parts.- Standard parts (MS or AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commerical utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided posse as suitable properties and are replaceable by the standard parts (MS or AN without alteration, and provided the corresponding standard part numbers are referenced in the parts list and if practicable, on the contractor' a drawings. In the event there is no suitable corresponding standand part in ef feet on date of invitation for bids, commercial part a may be used, provided they conform to all requirements of this specification.

3.5 Design.- Fuel float witches shall be designed in accordance with standard aircraft practices. The weight and sise of the unit shall be kept to a minimum. Reliability, however, shall be the paramount consideration l Pockets, wells, or traps, in which water or condensed moisture can collect shall be eliminated or properly drained. Consideration shall be given to the nece ssity for sufficient pretravel and overtravel of moving parts, and sufficient actuating force shall be available that wear and other deterioration of such parts shall not induce failure during the normal life of the component. A protective cover shall be provided to prevent damage to vital part a during shipment storage, Installation, and subsequent possible handling during maintenance and operation. Suffinient clearance shall be provided between the switch mechnaism and the cover to minimize the possiblity of malfunction in the event of damage to the protective cover. Consideration shall be given to the effect of icing] of the switch and float mechanism, and the switch shall be examined for icing effects during low temperature test.

3.5.1 Special tools.- When the float switch is a repairable item, disassembly and replacement of parts shall be accompli shable without the use of special tools.

3.5.2 Lubrication Float switches shall not require added lubrication between overhaul periods. Parts in contact with the fuel shall not require lubrication.

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3.5.3 Detailed design.- The detailed design requirements shall be specified in a detail specification or drawing and shall include the application and brief description of all the applicable data listed below. Where specific requirements of this specification are not applicable, the detail specification shall so state.

(a) Mounting (top, bottom, or side) _____
(b) Number of floats (single or multiple)_____
(c) Is relay required_____
(d) Data on relay _____
(e) Actuating levels (each float) _____
(f) Make or break contact on rising or lowering fuel level _____
(g) Operating current _____, volts (ac or dc)_____
(h) Maximum resistive load _____amps
(i) Maximum inductive load _____amps
(j) Inverted flight_____
(k) Precheck_____
(l) De-icing_____
(m) Negative "G" _____
(n) Weight _____
(o) Sketch or drawing indicating overall dimensions and mounting flanges_____

3.5.4 Electrica1.- Electrical devices shall conform to the applicable requirements of MIL-STD-704 and to the following requirements, when applicable.

3.504.1 Wire.- Internal electrical wire, other than windings, shall be In accordance with MIL-W-5086. External wire shall be in accordance with MIL-W-5088.

3.5.4.2 Connectors.- The electrical connectors shall be in accordance with MIL-C-26482 or MIL-C-26500. The size and type of connectors shall be as specified on the manufacturer's drawings. When necessary, leakproof connectors shall be employed.

3.5.4.3 Grounding. Each electrical connector used in the float switch shall incorporate a ground return terminal. Circuits shall not be Internally grounded.

3.5.4.4 Inductive loading.- The inductive electrical loading of dc electrical contacts during teat shall be provided through the use of an inductor in accordance with AN3179. The inductive electrical loading of ac electrical contacts during test shall be provided through the use of a power factor of 50 percent.

3.6 ~ * - The float switch shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, Installation, and service use. Any parts having relative motion to other parts shall be so constructed that binding, stickling, hanging, jamming, or other malfunction under operating and test conditions shall be positively prevented.

3.6.1 'O" rings All "O" rings shall conform to MIL-P-5315 and the dimensional requirements of Ms29513.

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3.6.2 Soldering. Soldering shall be adequately controlled to prevent possibility of direct damage to adjacent electrical parts. Flux used shall not induce corrosion.

3.7 Performance The float switch shall perform satiafaotorily when sub. jected to the tests specified in 4.6.

3.8 Interchangeable AU parts having the came manufacturer's part nnmber shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-70327 shall govern changes In the manufacturers part number.

3.9 Threaded connections

3.9.1 Pipe threads.- Pipe threads shall not be used on float switches, except for permanent closures. in which case the pipe threads shall conform to MIL-P-7105.

39992 Screw threads. Screw threads shall conform to MIL-S-7742

3.9.3 Locking of parts.- All threaded parts shall be looked by safety wiring, by all-metal self-locking nuts conforming to MIL-N-250Z7, cotter pins, or other approved methods. Safety wire shall be installed h accordance with MS33540 and shall conform to MS20995 , Self-locking nuts shall be used in accordance with MS33588. However, self-looking nuts shall not be used without specific authorization of the procuring activity where loosening or diaengagament of the self-looking nut could result in the nut or other part entering the fuel system. The use of lock washer or staking is not permitted.

3.9.4 Bosses and fitting ends.- Bosses and fitting ends shall conform to AND10049, AND10050, and MS3656, as applicable.

3•10 Temperature range.--- The float switch shall operate over a fuel temperature and ambient temperature range f rom -67° F to +158° F.

3.11 Pressure range.- The float switch shall operate over a pressure range from 30 psi to a pressure equivalent to 80,000 feet altitude.

3.12 ~ * - The float switch shall conform to the manufacturer's drawings, l s approved by the procuring activity.

3.13 Color identification.- Each float switch shall be marked gloss red conforming to Federal Standard No. 595, color 11105. The marking shall consist of a band 0.25 inch tide by 1 inch long on each aide or around the component, and a 45-degree arc 0.25 inch wide on both aides of this component. The color shall be parmanent, and when applied to parts that may come in contact with fuel, the color shall not soften, tip, crack, or cause the fuel to be affected in any way.

3.14 U.- Anodizing, plating other than cadmium or chromium and other protective treatment of float switches shall be as specified in the model specification and applicable detail drawings.

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3.14.1 Cadmium plating.- Cadmium plating shall conform to class 2, type II of QQ-P-416. Cadmium-plated parts shall not be used in contact with fuels, unlearn Specifically authorized by the procuring acotivity.

3.14.2 Chromium plating.- - Chromium plating shall conform to class 2 of QQ-C-320.

3.l4.3 Paint.-Paint shall not be used, except for color code identification as Specified in 3.13.

3.15 Synthetic rubber parts

3.15.1 Maintainsbility.- All synthetic rubber parts shall be readily replaceable with a minimum replacement of attaching parts.

3.15.2 Age controls.- Age controls shall conform to ANA bulletin No. 438.

3.16 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.16.1 Namplate.- A nameplate conforming to MIL-P-P4906 shall be attached securely to the float switch, or the information shall be etched, engraved, embossed, or stamped in a suitable location thereon. In addition to marking required by MIL-STD-130, the following shall be included:

Contractar's model specification number
Serial number

3.17 Report of preproduction tost.- The reports submitted in compliance with 4.3.1 shall include the following:

(a) Report of all tests, graphically presented when possible, together with a detailed statement indicating compliance or extent of noncompliance with all requirements of this specification, referring specifically to paragraph numbers. Wherever a requirement is considered to be inapplicable, the report should so state.

(b) Summary of endurance test.

(c) Diagrams of all test setups.

(d) Reproducible outline and description of tests and test conditions. Where tests specified In this specification are not considered applicable, the reason and the substituted test (if any) shall be described clearly.

(e) Copies of test log sheets.

(f) Photograph when available.

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3.18 Drawings.- Manufacturer's assembly and detail drawings shall conform to MIL-D-70327.

3.19 Workmanship.- All details of workmanship shall be sufficient to insure satisfactory operation and service life. Parts shall not contain sharp edges, burrs, loose chips, dirt, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- The inspection and testing of float switches shall be classified as:

(a)	Preproduction tests	(4.3)
(b)	Quality conformance tests	(4.4)

4.3 Preproduction tests.-

4.3.1 Sampling instructions.- Unless otherwise specified, each preproduction test sample shall consist of six switches of each manufacturer's part number upon which approval is desired. (Three untested float switches and three float switches that have been tested by the manufacturer in accordance with this specification prior to being forwarded to the testing facility.) The float switches shall be accompanied by one complete set of manufacturer's drawings, including any electrical wiring or circuit diagrams, the contractor's model specification, a parts list, and a complete test report showing the results of the contractor's tests. Samples shall be identified as required and forwarded to the testing facility designated in the contract (see 6.2).

4.3.2 Tests.- The preproduction tests of the float switch shall consist of the following tests, in the order listed, as specified under 4.6:

(a) Tests for Sample No. 1:

(1)	Examination of product	(4.6.1.1)
(2)	Weight	(4.6.19)
(3)	Calibration	(4.6.3)
(4)	Dielectric strength	(4.6.2)
(5)	Radio interference	(4.6.14)
(6)	Fungus resistance	(4.6.8)
(7)	Humidity	(4.6.9)
(8)	De-icing	(4.6.15)
(9)	Explosion-proof	(4.6.12)
(10)	Calibration	(4.6.3)
(11)	Leakage	(4.6.4)
(12)	Disassembly and inspection	(4.6.20)

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(b) Tests for Sample No. 2:

(1) Rumination of product	(4.6.1.1)
(2) weight	(4.6.19)
(3) Calibration	(4.6.3)
(4) Dielectric strength	(4.6.2)
(5) Immersion	(4.6.10)
(6) Fuel resistance	(4.6.18
(7) Endurance	(4.6.16)
(8)- Fluid dynamic loading	(4.6.17)
(9) Calibration	(4.6.3)
(10) Leakage	(4.6.4)
(11) Disassembly and inspecition	(4.6.20)

(c) Tests for Sample No. 3:

(1) Examination of product	(4.6.1.1)
(2) weight	(4.6.19)
(3) Calibration	(4.6.3)
(4) Dielectric strength	(4.6.2)
(5) Vibration	(4.6.7)
(6) Salt spray	(4.6.13)
(7) Mechanical shook	4.6.11)
(8) Proof pressure	(4.6.5)
(9) Burst pressure	(4.6.6)
(10) Disassembly and inspection	(4.6.20)

4.4 Quality conformance te sts.- Quality conformance tests shall consist of individual tests and sampling tests, 4.4.1 and 4.4.2, respectively.

4.4.1 Individual test.- Each float switch shall be subjected to the following tests, as specified under 4.6:

(a) Examination	(4.6.1)
(b) Dielectric strength	(4.6.2)
(c) Calibration	(4.6.3, 4.6.3.1, 4.6.3.6, and 4.6.3.7)
(d) Immersion	(4.6.10)

4.4.2 Sampling tests.- Sample float switches shall be selected at random from each inspection lot in accordance with the table titled "Sample size code letters" of MIL-STD-105, Inspection level S-2, acceptance number zero for all tests, and shall be subjected to the following teats, as specified under 4.6:

(a) Room temperature endurance	(4.6.16.1)
(b) Vibration	(4.6.7)
(c) Roof pressure	(4.6.5)
(d) Leakage	(4.6.4)
(e) Burst pressure	(4.6.6)
(f) Disassembly and inspection	(4.6.20)

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4.5 Test conditions

4.5.1 Atmospheric.- Unless otherwise specified, all teats required by this specification shalkl be at an atmospheric pressure of 28 to 32 inches of mercury and ● t a temperature of 60° to 90° F and ● relative humidity of 80 percent or less. Where teats are made uith atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading .

4.5.2 Altitude.- Altitude pressure *shall be* determined from U. S. Standard Atmosphere, 1962.

4.5.3 Attitude.- Unless otherwise specified, all tests shall be conducted with the component installed in the normal mounting attitude.

4.'5.4 Cleaning.- Prio to testing the float switch, all parts normally in contact with the fuel or fuel vapors shall be thoroughly cleaned to remove all lubricant 8, protective coatings, and foreign matter.

4.5.5 Test fluid.- Unless otherwise specified herein, the fluid used for testing shall conform to MIL-G-5572, MIL-J-5624, MIL-F-7024, or types I and III of TT-S -735, Other fluids, such as fluids conforming to P-D-680 may be wed only with the specific approval of the procuring activity.

4.5.6 Test pressure.- All pressures specified herein are gage pressures.

4.6 Test methods

4.6.1 Examinations

4 . 6 . 1 . 1 ~ . - Each float switch submitted for acceptance shall be carefully examined for material, size workmanship, and dimensional requirements to determine conformance to the requirement s of thi a specification,

4.6.1.2 Preservation, packaging, packing, and marking.- Preparation for delivery shall be inspected for conformance to section 5 of this specification.

4 . 6 . 2 ~ * - The applicable teat voltage as specified in table I shall be applied between circuits and between each circuit and the metal frame of the flo at switch. Electrical current flow in excess of 2 milliampers or breakdown *of* insulation shall consitute failure.

TABLE I. Test voltage

Normal voltage	Test voltage (peak)	Time
28V dc	1,000V	1 min. 1/
115/200V 400-cycle ac	1,500V	1 min. 1/

1/ For individual tests (see 4.4.1), this time may be reduced to 1 second.

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4.6.3 Calibration

4.6.3.1 w specific gravity.- The float switch shall be mounted in the normal installation attitude. Test fluid having a specific gravity of 0.68 ±0.02 shall be used. A calibration cycle shall be perormed consi sting of increasing and then decreasing the test fluid level and noting the two fluid levels at which switch actuation occurs. Float switch actuation levels shall be as specified on the manufacturer's drawing with the exception that each actuation point tolerance shall be ±0.25 inch. The calibration test shall be conducted using the maximum rated resi stive load on all ontacts. There shall be no evidence of sticking or binding. Float switches with relays shall be tested using minimum line voltage applied to the relay.

4 . 6 . 3 . 2 ~ * - The float switch shall be mounted in the normal installation attitude. Test fluid having a specific gravity of 0.85 ±0.02 shall be used. A claibration cycle shall be performed consisting of increasing end then decreasing the test fluid level and noting the two fluid levels at which switch ● otuation occurs. Float switch ● otuation levels shall be as specified on the manufacturer's drawing with the exception that each actuation point tolerance shall be ±0. 25 inch. The calibration test shall be conducted using the maximum rated reed stive load on all contacts. There shall be no evidence of sticking or binding. Float switches with relays shall be tested using minimum line voltage applied to the relay.

4 . 6 . 3 . 3 - - - The float switch shall be functionally tested in four positions 990 degrees apart, when tilted at an angle of 15 degrees. Tout fluid ving a specified gravity of 0.66 4).02 shall be used. The calibration test of ..6.3.1 shall be conducted. There shall be no evidence of sticking or binding.

4 , 6 . 3 . 4 ~ . - The float switch shall be Installed on a pressure vessel and positioned in the normal mounting attitude. This teat shall be conducted in the same manner as the low specific gravity test fluid calibration teat, except that pressure in the vessel shall be maintained at 30 pal. There shall be no evidence of sticking or binding.

4.6.3.5 Inverted.- The float switch in a dry condition shall be slowly rotated from a normally mounted ● ttitude to an inverted ● ttitude and returned to the normal attitude. This procedure shall be conducted in four directions 90 degrees apart. The actuation or non-actuation of the float switch shall be in ● ccordance uith the detail specification.

4.6.3.6 Precheck.- The float switch precheck device shall be functionally tested in accordance with the detail specifications. There shall be no evidence of sticking or biding.

4.6.3.7 De-icing.- The float switch de-icing device sall be functionally tested in accordance with the detail specification.

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4.6.4 Leakage.- The float switch shall be Installed on a pressure vessel and the vessel filled with teat fluid. The vessel shall be positioned in such manner that the float switch mounting seal is in contact with the test fluid at all times. A pressure Of 30 psi shall be applied to the test fluid, Any leakage occurring **irlth.in a** 10-minute waiting periods commencing with the application-of pressue, shall constitute ***failure.*** **Float switches designed for mounting within ● fuel tank shall also be subjected to 30 pal test fluid pressure for a 10-minute period. Any evidence of leakage into the sealed portions of the float switch, as disclosed by disassembly and inspection on** completion **of all tests, shall constitute failure.**

4.6.5 Proof pressure.- The float switch shall be installed on a pressure vessel and the vessel filled with test fluid. The vessel shall be positioned in such manner that the float switch mounting seal is in contact with the test fluid at all times. A pressure of 60 psi shall be applied to the teat **fluid for a 5-minute period. When the pressure is reduced to 30 psi,** there shall be no external leakage during **● 10-minute waiting period. There shall be no evidence *of*** failure, malfunction, ***or*** **distortion, and the low specific** gravity teat shall be performed in accordance with 4.6.3.1, This test also applies to float switches designed for mounting within fuel tanks.

4.6.6 Burst pressure.- The float switch shall be installed on a pressure vessel and the vessel filled with teat fluid. The vessel shall be positioned in such manner that the float switch mounting seal is in contact with the test fluid at all times. A pressure of 90 psi shall be applied to the test fluid **for a 5-minute period. When the pressure is reduced to 30 psi, there shall be no external leakage during** a 10-minute waiting period. The float switch shall be functionally operational when tested with the low specific gravity test as described in 4.6.3.1. This test also applies to float switches designed for mounting within fuel tanks.

4.6.7 Vibration.- The float switch shall be subjected to vibration test Prroce-dure XII of MIL-E-5272. Half of the total vibration period shall be conducted with the switch assembly in the dry state. The cycling times, resonant scanning, and prolonged vibration shall all be proportioned between a condition of operation and a condition of nonoperation. The maximum rated resistance load shall be contin-**uously** monitored by means of an oscilloscope for any erratic or inadvertent operation during vibration. Float switches with relays shall be tested, using minimum line voltage applied to the relay. The test fluid level shall be maintained within the actuation range of the float switch. Any erratic or inadvertent operation during teat or any evidence of failure or distortion shall constitute failure. On **completion of teat, the low specific gravity teat shall be performed in accordance with 4.6.3.1.**

4.6.8 Fungus resistance The float switch shall be **subjected to fungus resistance test Procedure I of MIL-E-5272. After completion of the test, the complete** float switch shall be subjected to the low specific gravity test in accordance with 4.6.3.1.

4.6.9 Humidity.- The float switch shall be subjected to humidity test Procedure I of MIL-E-5272. There shall be no evidence of malfunction or deterioration that might adversely affect operation. At the end of the test period, the low specific gravity test shall be performed in accordance with 4.6.3.1.

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4.6.10 Immersion.- **The float switch shall be subjected to Immersion (leakage) test Procedure I of MIL-E-5Z72. Leakage as defined in the test procedure shall[constitute fail-.** At **tbe** end of the test period, the low specific gravity test shall be performed in acccordance with **4.6.3.1**

4.6.11 Mechanical The float switch shall be subjected to equipment operation shock teat Procedure V of MIL-E-5272. Maximum line voltage shall be applied during test. At the end of this test, there shall be no evidence of failure or distortion, and the *low specific* gravity test shall be performed in accordance with 4.6.3.1,

4.6.12 Exposion.- The float switch shall be subjected to explosion-proof test Procedure III or IV, as applicable, of MIL-E-5272 The unit shall be calibrated on **completion of the teat in accordance with 4.6.3.1.**

4.6.13 ~ * - The float switch shall be subjected to salt spray test Procedure I of MIL-E-5Z72. At the end *of the test* period, the salt deposit shall be removed with warm **water, the swicth dried, and the *low* specific gravity test shall be performed In ❋ ccordance with 4.6.3.1. Any evidence of** corrosion **which might ●** chwrmly **● ffect the** float switch shall be ***cause for* rejection.**

4.6.14 ~ * - The float switch shall be subjected to a radio Interference test in accordance with MIL-STD-826.

4.6.15 De-icing.- Float switches with de-icing heaters shall be tested at maximum line voltage and in an ambient temperature of 160° F. The maximum temperature of heated parts in contact with fuel or fuel vapors shall be measured and shall not exceed **275° F.** The heated parts shall be submerged in fuel conforming to grade JP-4 of MU-J-5624, at **160°** F for 100 hours while maintaining maximum line voltage. At the end of the teat period, the low specific gravity test shall be performed in accordance with 4.6.3.1.

4.4.16 ~ * - The float switch shall be operated for a minimum of 100,000 complete cycles. The maximum rated resistive load shall be applied for 90,000 cycles and the maximum rated inductive load shall be applied for 10,000 cycle a. The endurance test shall consist of the following:

(a) Room temperature (70 percent of total).
(b) Altitude (I5 percent of total).
(c) Low temperature (5 percent of total).
(d) High temperature (5 percent of total).
(e) Contaminated fuel (5 percent of total).

The cycling shall be performed at a rate of no less than 5 cycles per minute. A cycle consists of float switc actuation on **both increasing and** decreasing fuel levels. Endurance testing may **be performed** either by raising or lowering the float **switch in the** test fluid or by clamping the float switch and raising and lowering the test fluid. Float switches with relays shall be tested using minimum **voltage applied to the relay.**

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4.6.16.1 Room temperature.- The room temperature endurance test shall be conducted et room temperature using a suitable test fluid. At the end of the test period, the low specific gravity test shall be performed in accordance with 4.6.3.1

4.6.16.2 Altitude.- The altitude endurance test shall be conducted at a simulated altitude of 80,000 feet, except that the ambient temperature shall be room temperature. Fuel conforming to grade JP-4 of MIL-J-624 shall be used and shall not be weathered. The low specific gravity test shall be performed at the beginning and end of the test period in accordance with 4.6.3.1.

4.6.16.3 Low temperature.- The low temperature endurance test shall be conducted using a fluid and ambient temperature of -67° ● 2° F. A suitable test fluid shall be used. At the end of the test period, the low specific gravity test shall be performed in accordance with 4.6.3.1.

4.6.16.4 **The high temperature endurance test shall be conducted, using a fluid and ambient temperature of 158° *2° F. A suitable test fluid shall be used. At the end of the teat period, the low specific gravity test shall be performed in accordance with *4.6.3.1.***

4.6.16.5 Contamination fuel.- **The contaminated fuel endurance test shall** be conducted at room temperature using a suitable test fluid and contaminant as specified in MIL-F-8615. The test fluid and contaminant shall be circulated. **At the end of the** test **period, the low** specific gravity test shall be performed in **accordance with *4.6.3.1***

4.6.17 Fluid dynamic loading - **That** portion of the float switch projecting into the fuel and subjected to fluid force as a result of fuel movement shall be tested as follows: A mechanically simulated fluid loading of 5 psi of projected area shall be alternately applied on opposite sides of the projecting member normal to the major axis at a rate of 12 cycles per minute for a 10-hour period. At the end of the test period, there shall be no evidence of failure or distortion and the low specific gravity test shell be performed ***in*** **accordance** with 4.6.3.1.

4.6.18 Fuel resistance and low temperature.- The float switch shall be subjected to the applicable fuel resistance and low temperature test as specified in MIL-E-8615

4.6.19 Weight.- The float switch shall be weighed in a dry condition. The weight shall not exceed the weight shown on the manufacturer's drawing.

4.6.20 Disassembly After completion of all preproduction tests, the float switch shall be disassembled and inspected. There shall be no evidence of failure, deterioration, or undue wear which might affect performance of the switch.

5. PREPARATION FOR DELIVERY

5.1 Float switches shall be prepared for delivery in accordance with MIL-P-7936. Preservation for level A shall be Method IA of MIL-P-116. Unit quantities shall be one each.

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5.2 Marking of shipments. In addition to any special marking required by the **contract or order, unit packages, intermediate** packages, and **shipping containers shall be marked in accordance with MIL-STD-129.**

6. NOTES

6.1 Intended use.- The aircraft fuel level float switch covered by this specification is used for the purposes ***of controlling the*** **operation of pumps, valves, or actuating warning lights, buzzers, alarms, or other signalling devices, It is intended to mount in or on aircraft fuel tanks using hydrocarbon fuel having an ambient temperature range from** ***-67° to*** **+158° *F and* ● pressure range from 30 pal to a pressure equivalent to 80,000 feet altitude.**

6.2 Ordering data.- **Procurement documents should specify:**

(a) Title, **number, and date of detail specification relating to components to be furnished.**

(b) **The requirements for ●** contractor's model **specification and preproduction sample (see 3.3 and 4.3.1).**

(c) Number *of* units required **for ● preproduction test sample; where the preproduction test samples should be sent; the activity** responsible **for testing and instructions concerning the sumittal of the test reports (see 4.3.1).**

(d) **Quantity required.**

(e) Applicable levels of preservation, packaging, and packing (see 5.1).

(f) ***If teat reports are*** to be submitted (see **3.17 and 4.3.1).**

6.3 Provisions Manufacturing of any additional components **on the particular contract should be suspended until the samples submitted for preproduction tests are pronounced satisfactory by the procuring ● etivlty. When ● contractor is in continuous production of these components from contract to contrast, submit sion of further** preproduction samples on the **subsequent contract s may be waived at the discretion of the procuring activity. Approval of preproduotion samples or the valving of preproduction tests does not preclude the requirements of ● aaeptanoe testing.**

6.3.1 It is understood that the component supplied under contract or order will be identical with the corresponding preproduction ● ample in design, construction, quality. material, workmanship, and method of manufacture. Evidences of unauthorized change may consitute cause for rejection. Deviation from the preproduction sample should be made only under ● dvanoe approval by the procuring activity.

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6.4 Marginal indicia. - The margins of this specification have been marked to indicate where changes, deletions, or additions to the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last ***previous issue.***

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-0378

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